Persona Based Information Retrieval

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1 Context

The goal of this project is to build a search engine that can find relevant answers to user questions from a non-factoid Question Answering dataset. The system combines traditional statistical methods with modern neural ranking models to improve ranking and presentation.

2 Project Overview

1. Why are BM25/TF-IDF models alone insufficient?

They rely only on counting how often each word appears and don't take into account what those words actually mean. As a result, they can't recognize synonyms or grasp the full context of a query, so they often miss relevant documents that use different phrasing.

2. The use of a neural retrieval model can capture deeper semantic relationships?

Neural retrieval models use deep learning to understand the meaning behind queries and documents, not just surface word matches. By converting text into context-aware embeddings, a neural retrieval model can spot subtle semantic links, like paraphrases or related concepts, and give more accurate relevance scores, improving the ordering of search results.

3. Can query expansion increase the quality of the final ranking?

Yes. By adding synonyms, related words, and clearer phrases to the original query, query expansion uncovers documents that use alternate terms. This brings in more relevant candidates and lets the ranking model sort out the best ones more accurately.

Expected results

4. Increased recall and precision compared to the baseline.

- **Query Expansion:** adding synonyms and related terms boosts recall significantly while precision rises moderately if noise is well-filtered.
- **Novice Reformulation:** using simple, common language yields a moderate recall gain and keeps precision stable thanks to clear phrasing.
- **Expert Reformulation:** injecting domain-specific terms maximizes precision and still improves recall, but the recall gain isn't as large as the one achieved through broad query expansion.

5. Improved nDCG scores thanks to the integration of post-expansion.

The nDCG scores may improve because query expansion can help rank truly relevant documents higher in the results list. By expanding the query and applying both lexical and neural retrieval methods, the system can better identify relevant documents and place them earlier in the ranking, which directly benefits nDCG.

3 Methodology

Summary

We used the **antique/test** dataset, consisting of 200 queries and 403,666 documents.
The complete document collection was used as the searchable corpus. The qrels dataframe was not used during corpus construction or retrieval, but only during the evaluation stage, in order to avoid data leakage.

Data loading and cleaning

- Loading *queries* dataframe
- Loading *docs* dataframe
- Loading *qrels* dataframe

Data exploration

- Query response distribution
- Distribution of response lengths (in words)
- Distribution of query lengths (in words)
- Top word frequency in responses (excluding stop words)
- Top word frequency in queries (excluding stop words)

Phase I

- Baseline retrieval (BM25)
- Neural retrieval
- Functions section: evaluation metrics – precision, recall, MAP, nDCG
- Evaluation metrics
  - BM25: dataset preparation, precision and recall, MAP, nDCG
  - Neural retrieval: dataset preparation, precision and recall, MAP, nDCG

Phase II

Query expansion

- BM25
- Neural retrieval
- Evaluation metrics
  - BM25: dataset preparation, precision and recall, MAP, nDCG
  - Neural retrieval: dataset preparation, precision and recall, MAP, nDCG

Query reformulation – novice

- BM25
- Neural retrieval
- Evaluation metrics
  - BM25: dataset preparation, precision and recall, MAP, nDCG
  - Neural retrieval: dataset preparation, precision and recall, MAP, nDCG

Query reformulation – expert

- BM25
- Neural retrieval
- Evaluation metrics
  - BM25: dataset preparation, precision and recall, MAP, nDCG
  - Neural retrieval: dataset preparation, precision and recall, MAP, nDCG

Evaluation metrics
Precision, Recall and MAP were computed using binary relevance judgments, where relevance levels 1 and 2 were considered non-relevant and relevance levels 3 and 4 were considered relevant. nDCG was computed using the original graded relevance judgments, preserving the different relevance levels.
For all metrics, k represents the number of retrieved documents considered for each query.

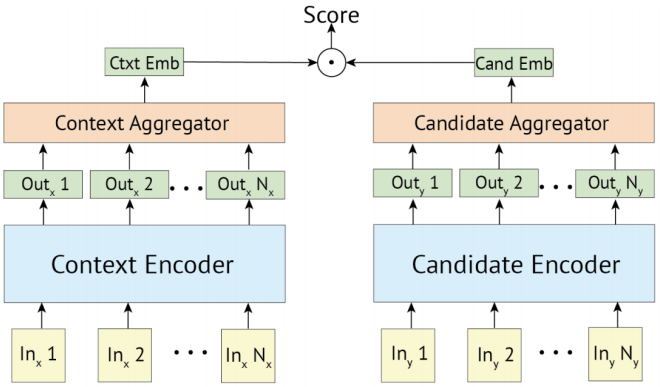
BM25

BM25 (Best Matching 25) is a ranking algorithm used in information retrieval to evaluate the relevance of a document with respect to a query. It belongs to the family of probabilistic models and is based on the frequency of terms in both the documents and the query.

BM25 penalizes overly frequent terms, preventing common words from dominating the score, and adjusts for document length, so that longer texts are not unfairly favored simply because they contain more words.
It is widely used in search engines and information retrieval systems as an effective and fast baseline, thanks to its simplicity and strong performance.

Neural Retrieval (Bi-Encoder)

A Bi-Encoder is a model that turns a query and a document into separate vectors, then compares them to see how similar they are, in our case using cosine similarity. The query and document are encoded independently, so document vectors can be precomputed and stored. For each query, cosine similarity is computed against the document embeddings and the top-k documents are retrieved.



(a) Bi-encoder

Gemini

For the automatic generation of the expansion, novice and expert queries, the Gemini 2.0 Flash Lite model was used, a free language model developed by Google. It was chosen for its speed and ability to produce consistent and contextually relevant output, allowing different levels of expertise to be simulated in the generated queries.

4 Results

BM25 Scores

Dataset	Precision@5	Recall@5	MAP@5	nDCG@5
original_dataset	0.24	0.11	0.07	0.47
expanded_dataset	0.15	0.08	0.06	0.23
novice_dataset	0.11	0.05	0.04	0.19
expert_dataset	0.10	0.06	0.04	0.17

Neural Retrieval Scores

Dataset	Precision@5	Recall@5	MAP@5	nDCG@5
original_dataset	0.43	0.21	0.18	0.51
expanded_dataset	0.37	0.19	0.16	0.44
novice_dataset	0.26	0.13	0.10	0.32
expert_dataset	0.30	0.16	0.13	0.35